

ZHEN LENG THAI

Large Language Model (LLM) Algorithm Engineer

Cheras, Kuala Lumpur, Malaysia | +60-182870172 | thaizhenleng123@gmail.com

EDUCATION

Tsinghua University	Sep 2024 – June 2026
M.S. in Computer Science and Technology GPA: 3.95/4.0 Awards: Chinese Government Scholarship	
Tsinghua University	Sep 2020 – June 2024
B.S. in Computer Science and Technology GPA: 3.85/4.0 Awards: Outstanding Graduate Award	

WORK EXPERIENCES

ModelBest Inc.	LLM Algorithm Engineer Intern	Feb 2024 – Feb 2026
- Architected MiniCPM-SALA hybrid attention, integrating sparse/linear mechanisms for 3.5x faster inference and a 2M+ context window. - Independently engineered a proprietary tokenizer for the MiniCPM series, optimizing data compression logic for enhanced multi-scenario processing efficiency. - Optimized hyperparameter frameworks and scaling laws to improve the predictive capabilities of small models over large model performance.		
Tencent Technology Co.	LLM Algorithm Engineer Intern	June 2023 – Sep 2023
Led the post-training of Agent capabilities, reinforcing intent understanding and backend tool utilization to drive user operation automation.		

RELEVANT PROJECTS:

Published 10 core papers with 2200+ Google Scholar Citations and an h-index of 5.		
Hybrid Linear Attention Done Right		Co-First Author
- Proposed the first high-performance hybrid attention architecture (25% sparse, 75% linear) deployed in MiniCPM-SALA, achieving 3.5X inference acceleration over Qwen3-8B while maintaining excellent length generalization - Designed a rapid three-stage Transformer conversion algorithm preserving core retrieval layers, and integrated HyPE position encoding to expand the context window to over 1 million tokens.		
Between a Rock and a Hard Place: Ethical Reasoning and Safety Alignment		Second Author
- Proposed the TRIAL multi-turn red-teaming framework to reveal binary safety alignment failures in complex moral reasoning systematically. - Constructed the ERR defense framework, utilizing layered Harm-Gated LoRA to robustly defend against reasoning-based attacks.		
DecorateLM: Data Engineering for Corpora via LMs		Co-First Author
EMNLP 2024 Main: Proposed a fine-grained data engineering pipeline (rating/tagging/editing) to significantly enhance pre-training data quality and diversity. - Trained a 1.2B proxy model on the optimized data, notably boosting large model input paradigms and generalization across reasoning and diversity tasks.		
Cost Optimal Grouped-Query Attention for Long-Context Modeling		Third Author
EMNLP 2025 Main: Decoupled head and hidden sizes within the default GQA configuration, reducing memory and FLOPs by >50% without performance loss in long-context scenarios.		
MiniCPM: Unveiling the Potential of Small LMs		Core Team Member
COLM 2024 (8.6k GitHub Stars): Developed the WSD learning rate scheduler, enabling sub-4B models (MiniCPM-2B) to rival the performance of significantly larger models (e.g., Llama2-8B).		

SKILLS & LANGUAGES

- Technical Skills:** Natural Language Processing (NLP), Model Architecture, Long-context Modeling
- Languages:** Chinese (Fluent), English (Fluent, IELTS 7.5), Malay (Fluent), Cantonese (Proficient).